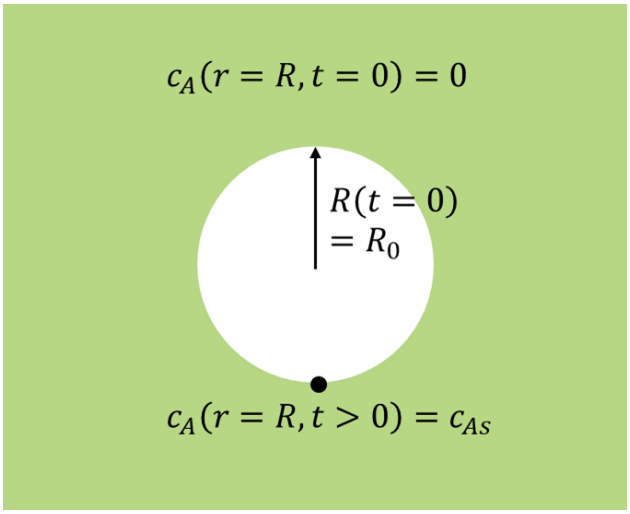
CBE 150B F2026
Problem Set 3 - Key

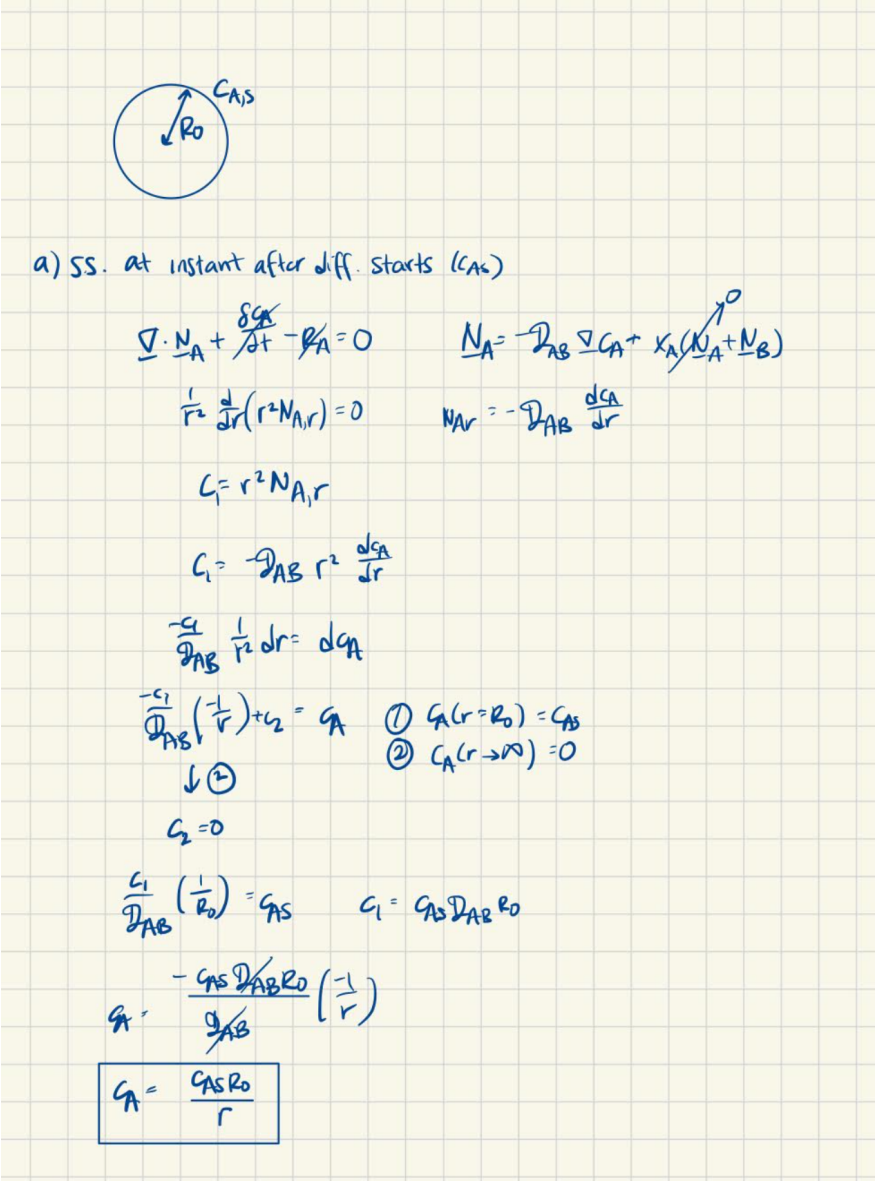
Assigned: September 9, 2026
Due: Septermber 16, 2026

Problem 1 (25 pts)

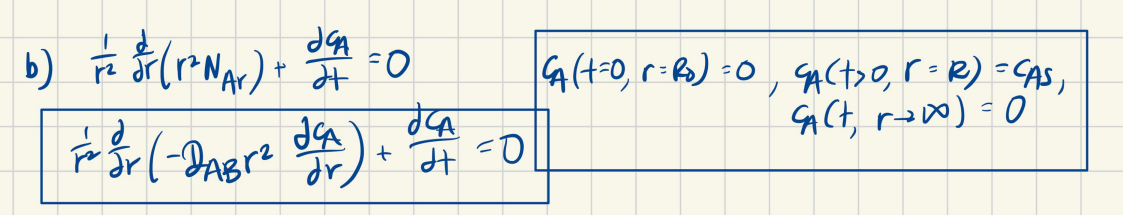
A spherical sugar pellet is dissolving in your infinitely large teacup. Initially, the pellet is of radius R_0, and there is no dissolved sugar in the teacup. After this initial instant ($t > 0$), the surface of the spherical sugar pellet has an equilibrium concentration, C_{As}. The sugar has a diffusivity in tea of D_{AB}. Assume C_{As} is relatively dilute.



a) Derive the steady-state concentration profile for the initial instant at which the surface concentration reaches C_{As} and the radius is still R_0.

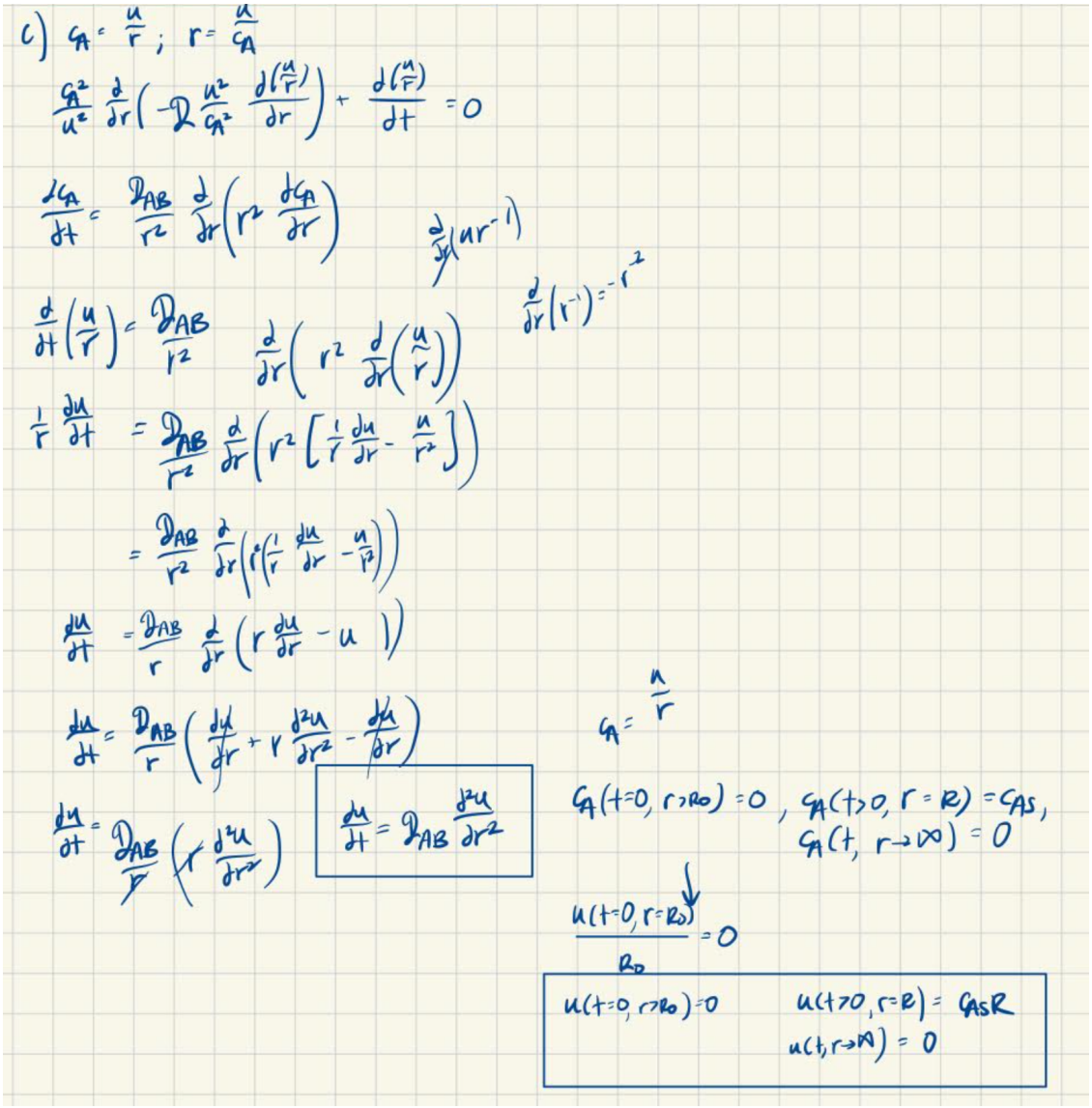


b) Write the differential equation governing the time-dependent diffusion of sugar (in terms of C_A, t, r). Include the initial condition and boundary conditions for this scenario.



Now, introduce the substitution $u = c_A \cdot r$ to eliminate C_A from your PDE.

c) With this substitution, simplify the PDE from (b). Write the new initial and boundary conditions with this substitution.



d) Solve the simplified PDE and write the resulting concentration profile for C_A as a function of r and t.

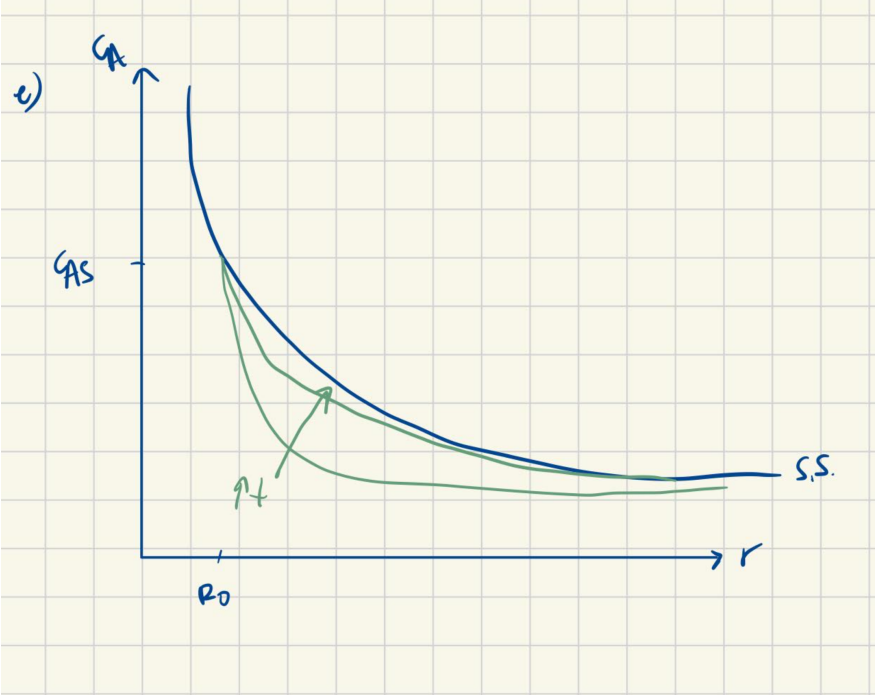
d) Shift origin to apply sln: $r^* = r - R_0$

$u(0, r^*>0) = 0, \quad u(t, r^*=0) = C_{AS} R_0, \quad u(t, r^* \to \infty) = 0$

$u = u(r^*=0)\left[1 - \operatorname{erf}\left(\frac{r^*}{\sqrt{4 D_{AB} t}}\right)\right] \Rightarrow \boxed{C_A = C_{AS} \frac{R_0}{r}\left[1 - \operatorname{erf}\left(\frac{r - R_0}{2\sqrt{D_{AB} t}}\right)\right]}$

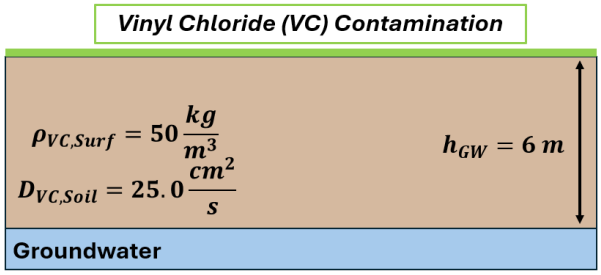
S.S. transient

e) Sketch the steady state concentration profile. On the same axes, sketch the time-dependent concentration profile at two times $t_1,\ t_2 > 0$.

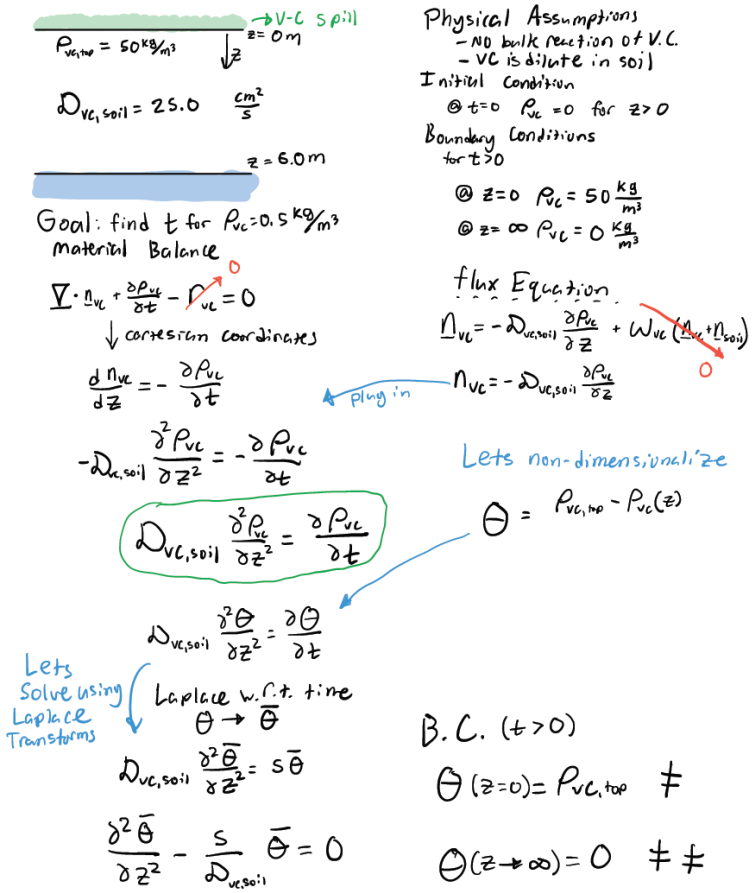


Problem 2 (20 pts)

In February 2023, a freight train carrying the monomer vinyl chloride (VC) derailed and crashed in Ohio. The crash resulted in a chemical spill of vinyl chloride across the ground, which posed a threat to local groundwater. Imagine you are part of the EPA's emergency response team, dispatched to the clean up. As the team's chemical engineer, you have been asked to track the spread of VC, and the threat it poses to the water table. You have surveyed the land and discovered that the water table sits 6.0 m below the surface. Vinyl chloride has saturated the topsoil with a concentration of $50\frac{kg}{m^3}$. The diffusivity of VC through soil is $D_{VC,Soil} = 25.0\frac{cm^2}{s}$. Your team considers VC concentrations of $0.5\frac{kg}{m^3}$ to be dangerous to the water table.



a) How much time (hours) do you have between the initial spill time and "breakthrough" of VC to dangerous levels at the water table.



Solve for general solution

$$\bar{\theta} = A_1 e^{\sqrt{s/D_{vc,soil}}z} + B_1 e^{-\sqrt{s/D_{vc,soil}}z}$$

Apply B.C. ##

as $z \to \infty$ $\bar{\theta} \to 0$ so $A_1 = 0$

Apply B.C. #

at $z = 0$ $\bar{\theta} \to \frac{\rho_{vc,surf}}{s}$ so $B_1 = \frac{-\rho_{vc}}{s}$

$$\therefore \bar{\theta} = \frac{\rho_{vc}}{s} e^{-\sqrt{s/D_{vc,soil}}z}$$

(Inverse La Place transform

$$\theta = \rho_{vc,surf} \operatorname{erfc}\left(\frac{z}{2\sqrt{D_{vc,soil}t}}\right)$$

$$\frac{\rho_{vc,surf} - \rho_{vc}}{\rho_{vc,surf}} = \operatorname{erfc}\left(\frac{z}{2\sqrt{D_{vc,soil}t}}\right) = 1 - \operatorname{erf}\left(\frac{z}{2\sqrt{D_{vc,soil}t}}\right)$$

Find t for $\rho_{VC,water} = 0.5 \frac{kg}{m^3}$

$z = 6.0\ m$ $\quad D_{O_2,N_2} = 25.0 \frac{cm^2}{s} = 2.50 \times 10^{-3} \frac{m^2}{s}$

$\rho_{VC} = 0.5 \frac{kg}{m^2}$ $\quad \rho_{VC,surf} = 50$

$$\frac{50 - 0.5}{50} = 1 - erf\left(\frac{6.0}{2\sqrt{(2.50\times10^{-3}\frac{m^2}{s})t}}\right)$$

$$\frac{\rho_{VC,surf} - \rho_{VC}}{\rho_{VC,surf}} = 1 - erf\left(\frac{z}{2\sqrt{D_{VC,soil}\,t}}\right)$$

$$erf\left(\frac{z}{2\sqrt{D_{VC,soil}\,t}}\right) = \frac{\rho_{VC,surf}}{\rho_{VC,surf}} - \frac{\rho_{VC,surf} - \rho_{VC}}{\rho_{VC,surf}} = \frac{\rho_{VC}}{\rho_{VC,surf}}$$

$$\rho_{VC}(z) = \rho_{VC,surf}\, erf\left(\frac{z}{2\sqrt{D_{VC,soil}\,t}}\right)$$

$\Downarrow$ plug in values

$$0.5\frac{kg}{m^3} = 50\frac{kg}{m^3}\, erf\left(\frac{6\ m}{2\sqrt{(0.00025\frac{m^2}{s})t}}\right)$$

$\Downarrow$ Desmos

$t = 4.58 \times 10^8$ seconds $\Rightarrow$ 1.27×10^5 hours

not much concern for the water table!

b) Your team plans to remove contaminated dirt from the crash site exactly two days (48 *hours*) after the crash. You want to remove any dirt contaminated with more than $5\frac{kg}{m^3}$ of VC. How many feet of soil must your team remove?

b. goal: remove soil in 48 hours w $\rho_{VC} \geq 5 \frac{kg}{m^3}$
what depth of soil must be removed?

$$\rho_{VC}(z) = \rho_{VC,surf}\, erf\left(\frac{z}{2\sqrt{D_{VC,surf}\, t}}\right)$$

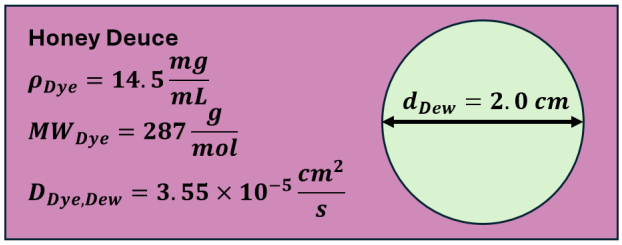
Find z for $\rho_{VC} = 5.0 \frac{kg}{m^3}$
$t = 48\,h = 172{,}800$ seconds

$$5.0 = 50.0\, erf\left(\frac{z}{2\sqrt{(0.00025 \frac{m^2}{s})(172800\,s)}}\right)$$

$z = 1.17$ m

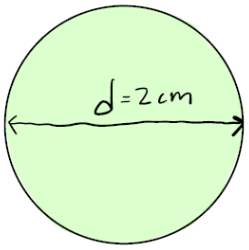
Problem 3 (15 pts)

The Honey Deuce is the official cocktail of the U.S. Open Tennis Tournament. It is a vodka-lemonade-based cocktail with the signature addition of raspberry liqueur. It is traditionally garnished with small spheres of honeydew melon. As honeydew spheres bob around in the cocktail, they soak up the red dye cyanidin ($MW = 287\frac{g}{mol}$) which gives raspberries their color. The melon spheres have a diameter of 2 cm, the cocktail remains on ice at $T = 0°C$, and raspberry coloring is present in the cocktail at a density of 14.5 $\frac{mg}{mL}$. Assume the cocktail remains well mixed with constant density of dye. The diffusivity of cyanidin through the honeydew melon is $D_{Dye,Dew} = 3.55 \times 10^{-5}\frac{cm^2}{s}$.



a) Does the diffusion of dye in this system take place in the dilute or concentrated regime? You may assume the cocktail has a similar molar density as water (What is the molarity of water?).

3: Honey dwece



$\rho_{dye} = 14.5\ \frac{mg}{mL}$

$MW_{dye} = 287\ g/mol$

$D_{Dye,Dew} = 3.55 \times 10^{-5}\ \frac{cm^2}{s}$

$T = 0°C$, $P = 1\ atm$

Physical Assumptions
- well-mixed cocktail
- ρ_{dye} is constant everywhere
- no reaction of dye

a. does dye diffusion of dye take place in dilute or concentrated regime?

Assume cocktail $\approx H_2O$

Find water concentration

$\rho_{H_2O,0°C} \approx 1.00\ g/cm^3$

$C_{H_2O} = \frac{\rho_{H_2O}}{MW_{H_2O}} = \frac{1.00\ g/mL}{18.0\ g/mol} \times \frac{1000\ mL}{1\ L} = 55.56\ M$

$C_{dye} = \frac{\rho_{dye}}{MW_{dye}} = \frac{0.0145\ g/mL}{287\ g/mol} \times \frac{1000\ mL}{1\ L} = 0.505\ M$

$\chi_{dye} = \frac{C_{dye}}{C_{total}} = \frac{0.505\ M}{(55.56 + 0.505\ M)} = 0.009$ → 0.9 mol% dye

Dilute Regime

b) How long would it take (in minutes) for the concentration of dye at the center of the melon sphere to reach 90 % of the dye concentration at the edge?

b. How many minutes for $C_{dye}(r=0)$ to be 90% of $C_{dye}(r=R)$

Recognize that we are dealing with a non-∞ solid, and our species is dilute so we can use the correlations in Appendix F. specifically figure F3 for spherical coordinates

Next, we need to identify our non-dimensionalized relative parameters

General → Our System → Specific Values

"Unaccomplished Change" $Y = \frac{C_{A1} - C_A}{C_{A1} - C_{A,0}} \Rightarrow \frac{C_{dye}(R) - C_{dye}(r)}{C_{dye}(R) - C_{dye,0}(r)} = \frac{0.505 - (0.9)(0.505)}{0.505 - 0} = 0.10$

"relative" Time $X = \frac{D_{AB}t}{X_1^2} \Rightarrow \frac{(D_{dye,dew})\,t}{R^2} = \frac{(3.55\times10^{-5}\,\frac{cm^2}{s})\,t}{(1.0\ cm)^2} = (3.55\times10^{-5}\,s^{-1})t$

"relative" Position $n = \frac{X(r)}{X(R)} \Rightarrow \frac{r_{center}}{R_{dew}} = \frac{0\ cm}{1.0\ cm} = 0$

"relative" Resistence $m = \frac{D_{AB}}{k_c X_1} \Rightarrow \frac{D_{dye,dew}}{k_{c,cocktail} R_{dew}} \Rightarrow 0$ b/c $k_c >>$

Using Figure F.3, find that $X = 0.35$

$0.35 = (3.55\times10^{-5}\,s^{-1})t \Rightarrow t = \frac{0.35}{3.55\times10^{-5}\,s^{-1}} = 9,860\,s$

$t = 9860\,s = 164$ minutes

c) Now, for this part ONLY, presume the cocktail does not remain well-mixed. In fact you determine there is a mass transfer coefficient of $k_c = 7.10 \times 10^{-5} \frac{cm}{s}$. How long will it take (in minutes) for the concentration at the melon's center to reach 90% of the edge concentration?

C. what if cock tail is no longer well stirred + $k_c = 7.10 \times 10^{-5} \frac{cm}{s}$

Y, X, n are the same

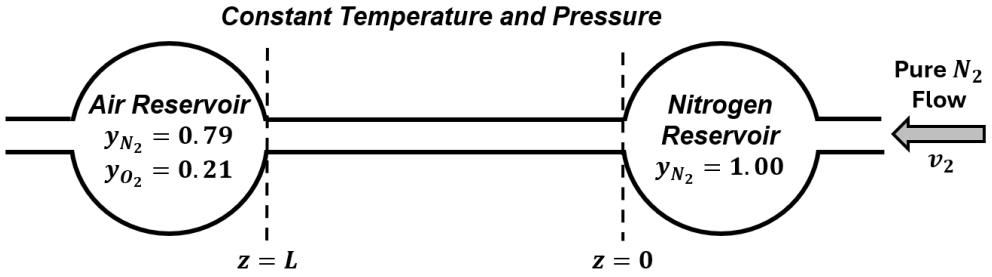
$$m = \frac{D_{AB}}{k_c X_1} \Rightarrow \frac{D_{edge,dew}}{k_{c,cocktail} R_{dew}} = \frac{3.55 \times 10^{-5} \frac{cm^2}{s}}{7.10 \times 10^{-5} \frac{cm}{s} (1.0 cm)} = 0.5$$

Using the new value of m on Figure F.3 $\Rightarrow X = 1.05$

$$t = \frac{1.05}{3.55 \times 10^{-5} s^{-2}} = 29,580 \text{ s} = 423 \text{ minutes}$$

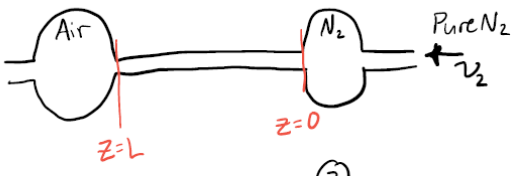
Problem 4 (15 pts)

Consider the problem of oxygen exclusion from a vessel at constant T and P, as covered in lecture Example 3. However, in this problem you should consider a case where the right reservoir is still pure N_2, while the left reservoir is air (79 mol % N_2, 21 mol % O_2), rather than pure O_2.



a) Derive the steady-state concentration profile for these conditions

Problem 5 - oxygen exclusion



① $y_{O_2,1} = 0.21$, $y_{N_2,1} = 0.79$

② $y_{N_2,2} = 1.0$

Physical Assumptions
- Line has constant cross sectional Area A
- Constant total Conc.
- constant T + P
- Steady State
  - no rxn
  - no accum
- 1-D cartesian coords

★ Boundary Conditions ★

@ $z = 0$, $y_{N_2} = 1$, $y_{O_2} = 0$

@ $z = L$, $y_{N_2} = 0.79$, $y_{O_2} = 0.21$

$N_{O_2} + N_{N_2}$ are constant

Component Balance

O_2: $\nabla \cdot \underline{N}_{O_2} + \frac{\partial C_{O_2}}{\partial t} - R_{O_2} = 0 = \frac{d N_{O_2}}{d z}$

N_2: $\nabla \cdot \underline{N}_{N_2} + \frac{\partial C_{N_2}}{\partial t} - R_{O_2} = 0 = \frac{d N_{N_2}}{d z}$

Component Fluxes ⇒ recall $N_{tot} = C_0 v_z = N_{O_2} + N_{N_2}$

O_2: $N_{O_2} = -C_0 D_{O_2,N_2} \frac{d y_{O_2}}{d z} + y_{O_2} \overbrace{C_0 v_z}^{N_{tot}}$

N_2: $N_{N_2} = -C_0 D_{N_2,O_2} \frac{d y_{N_2}}{dz} + y_{N_2} C_0 v_z$

$N_{O_2} = K_1 = -C_0 D_{O_2,N_2} \frac{d y_{O_2}}{d z} + y_{O_2} C_0 v_z$

$\frac{d y_{O_2}}{d z} - \frac{v_z}{D_{O_2,N_2}} y_{O_2} = \frac{K_1}{C_0 D_{O_2,N_2}}$

↳ solve Above for homogeneous 1st then Particular

Homog: $\frac{d y_{O_2}}{d z} - \frac{v_z}{D_{O_2,N_2}} y_{O_2} = 0$

$\frac{1}{y_{O_2}} d y_{O_2} = \frac{v_z}{D_{O_2,N_2}} d z$

$\ln(y_{O_2}) = \frac{v_z}{D_{O_2,N_2}} z + K_2$

$y_{O_2,homog} = K_3 \exp\left(\frac{v_z}{D_{O_2,N_2}} z\right)$

Particular:

$$y_{O_2} = y_{O_2,homog} + y_{O_2,part} = K_3 \exp\left(\frac{v_z}{D_{O_2,N_2}} z\right) + \frac{K_1}{C_0 D_{O_2,N_2}}$$

B.C. 1 $\quad y_{O_2} = 0 \quad @ \; z = 0$

$$0 = K_3 + \frac{K_1}{C_0 D_{O_2,N_2}}$$

$$K_3 = -\frac{K_1}{C_0 D_{O_2,N_2}}$$

$$K_1 = -C_0 D_{O_2,N_2} K_3$$

$$y_{O_2} = K_3 \left(\exp\left(\frac{v_z}{D_{O_2,N_2}} z\right) - 1\right)$$

BC #2 $\quad @ \; z = L \quad y_{O_2} = 0.21$

$$K_3 = \frac{0.21}{\left(\exp\left(\frac{v_z}{D_{O_2,N_2}} L\right) - 1\right)}$$

Therefore, bringing everything together

$$y_{O_2} = 0.21 \frac{\exp\left(\frac{v_z}{D_{O_2,N_2}} z\right) - 1}{\exp\left(\frac{v_z}{D_{O_2,N_2}} L\right) - 1} = 0.21 \frac{\exp\left(Pe_L \frac{z}{L}\right) - 1}{\exp(Pe_L) - 1}$$

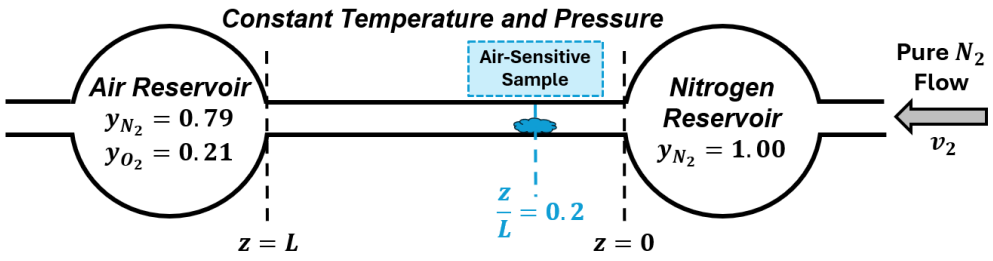
$$y_{O_2} = 0.21 \frac{\exp\left(Pe_L \frac{z}{L}\right) - 1}{\exp(Pe_L) - 1}$$

$$y_{N_2} = 1 - y_{O_2} = 0.79 \frac{\exp\left(Pe_L \frac{z}{L}\right) - 1}{\exp(Pe_L) - 1}$$

$$C_{O_2} = C_0 y_{O_2} = 0.21 C_0 \frac{\exp\left(Pe_L \frac{z}{L}\right) - 1}{\exp(Pe_L) - 1}$$

$$C_{N_2} = C_0 y_{N_2} = 0.79 C_0 \frac{\exp\left(Pe_L \frac{z}{L}\right) - 1}{\exp(Pe_L) - 1}$$

Now, consider that you are working in a laboratory with a precious air-sensitive sample. You are storing this sample in the line between the two reservoirs at the position $\frac{z}{L} = 0.2$.



b) What is the value of Pe_L required to maintain a steady-state mole fraction of oxygen $y_{O_2} = 0.001$ at $\frac{z}{L} = 0.2$

b. Air-sensitive sample @ $\frac{z}{L} = 0.2$, what Pe_L is needed to maintain $y_{O_2} = 0.001$?

$$y_{O_2} = 0.21 \frac{\exp(Pe_L \frac{z}{L}) - 1}{\exp(Pe_L) - 1}$$

$$0.001 = 0.21 \frac{\exp(0.2\, Pe_L) - 1}{\exp(Pe_L) - 1}$$

↓ Desmos

$$Pe_L = 6.27$$

c) What value of Pe_L is needed to maintain a steady state oxygen mole fraction of $y_{O_2} = 0.0001$ at $\frac{z}{L} = 0.2$ – ten times lower?

c. Air-sensitive sample @ $\frac{z}{L} = 0.2$, what Pe_L is needed to maintain $y_{O_2} = 0.0001$

$$y_{O_2} = 0.21 \frac{\exp(Pe_L \frac{z}{L}) - 1}{\exp(Pe_L) - 1}$$

$$0.0001 = 0.21 \frac{\exp(0.2\, Pe_L) - 1}{\exp(Pe_L) - 1}$$

↓ Desmos

$$Pe_L = 9.35$$

d) Based on your answers in (b) and (c), is there a linear scaling between nitrogen purity at the sample's location and N_2 velocity? In other words, for y_{O_2} to be reduced by a factor of 10, does nitrogen velocity need to increase by a factor of 10?

d. How does v_2 scale w/ concentration @ $\frac{z}{L} = 0.2$

for $y = 0.001$ $Pe_L = 6.27$

for $y = 0.0001$ $Pe_L = 9.35$

→ 10× change in y_{O_2}

$$\frac{Pe_{L,0.001}}{Pe_{L,0.0001}} = \frac{\frac{v_{2,0.001}}{D_{O_2,N_2} L}}{\frac{v_{2,0.0001}}{D_{O_2,N_2} L}} = \frac{v_{2,0.001}}{v_{2,0.0001}} = \frac{6.27}{9.35} = 0.671$$

$\frac{y_{O_2,0.001}}{y_{O_2,0.0001}} = 10$ but $\frac{v_{2,0.001}}{v_{2,0.0001}} = 0.671$

$\therefore$ Scaling is Not Linear

e) How would your answers to (b) and (c) change if the precious air-sensitive sample was stored at $\frac{z}{L} = 0.6$?

e.

If the sample is at $\frac{z}{L} = 0.6$, Pe_L & thus v_2 will need to increase